$$\frac{d}{dt}\langle \mathbf{P} \rangle = -\langle \frac{dV}{dx} \rangle.$$

Now, let's prove Eq. 9.16. Letting the commutator act on a wave function, we can write

$$[\mathbf{V}(x), \mathbf{P}]\psi(x) = V(x)(-i\hbar\frac{d}{dx})\psi(x) - (-i\hbar\frac{d}{dx})\,V(x)\psi(x). \tag{9.17}$$

This is easily simplified and results in Eq. 9.16. Thus, we have shown that

$$\frac{d}{dt}\langle \mathbf{P} \rangle = -\langle \frac{dV}{dx} \rangle, \tag{9.18}$$

which is the quantum analog of Newton's equation for the time rate of change of momentum.

Exercise 9.3: Show that the right-hand side of Eq. 9.17 simplifies to the right-hand side of Eq. 9.16. *Hint*: First expand the second term by taking the derivative of the product. Then look for cancellations.

9.7 LINEAR MOTION AND THE CLASSICAL LIMIT

You might think we have proved that the expectation value of **X** exactly follows the classical trajectory. But what we've actually proved is quite different. This difference exists because the average of a function of *x* is not the same as the function of the average of *x*. If Eq. 9.18 had read

$$\frac{d}{dt}\langle \mathbf{P} \rangle = -\frac{dV(\langle x \rangle)}{d\langle x \rangle} \quad [This\ is\ wrong]$$